RePr: Improved Training of Convolutional Filters

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Outline

1. Background
2. Related Work
3. Motivation

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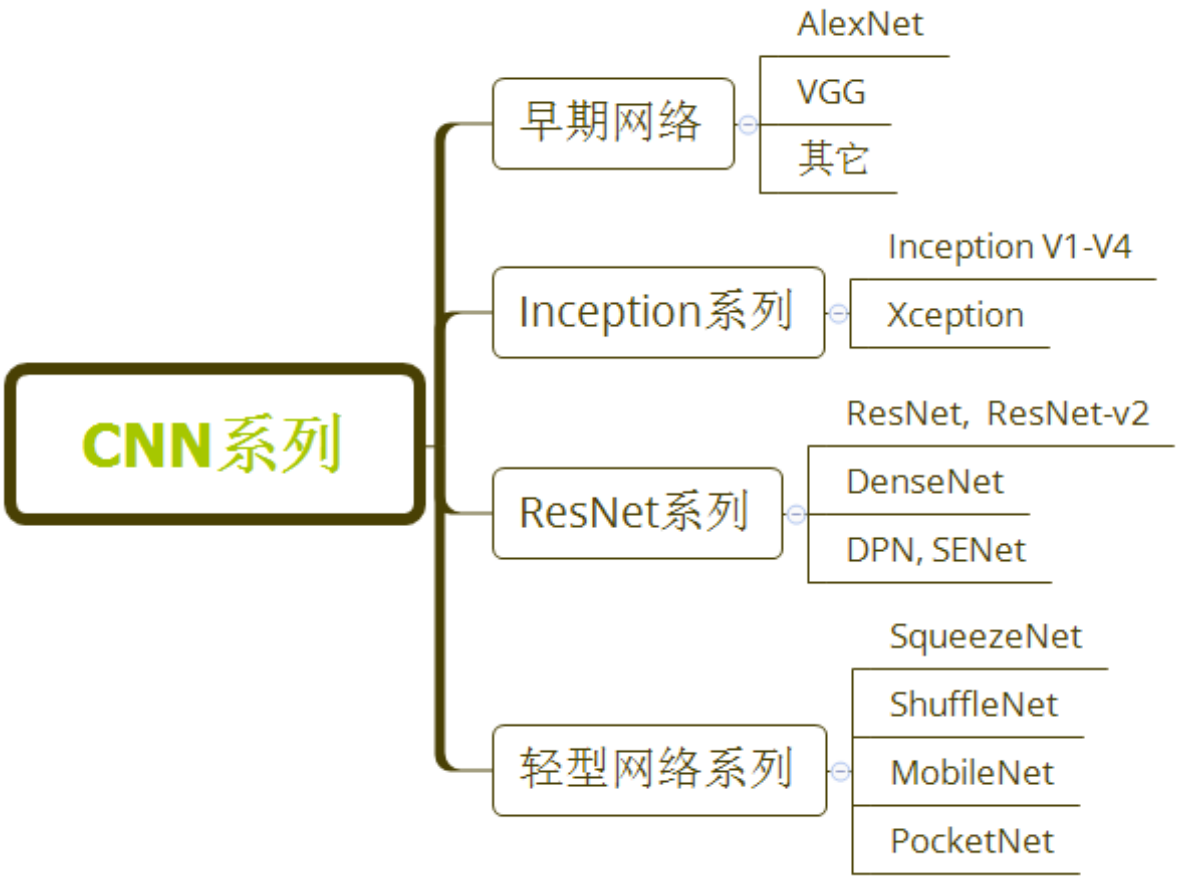
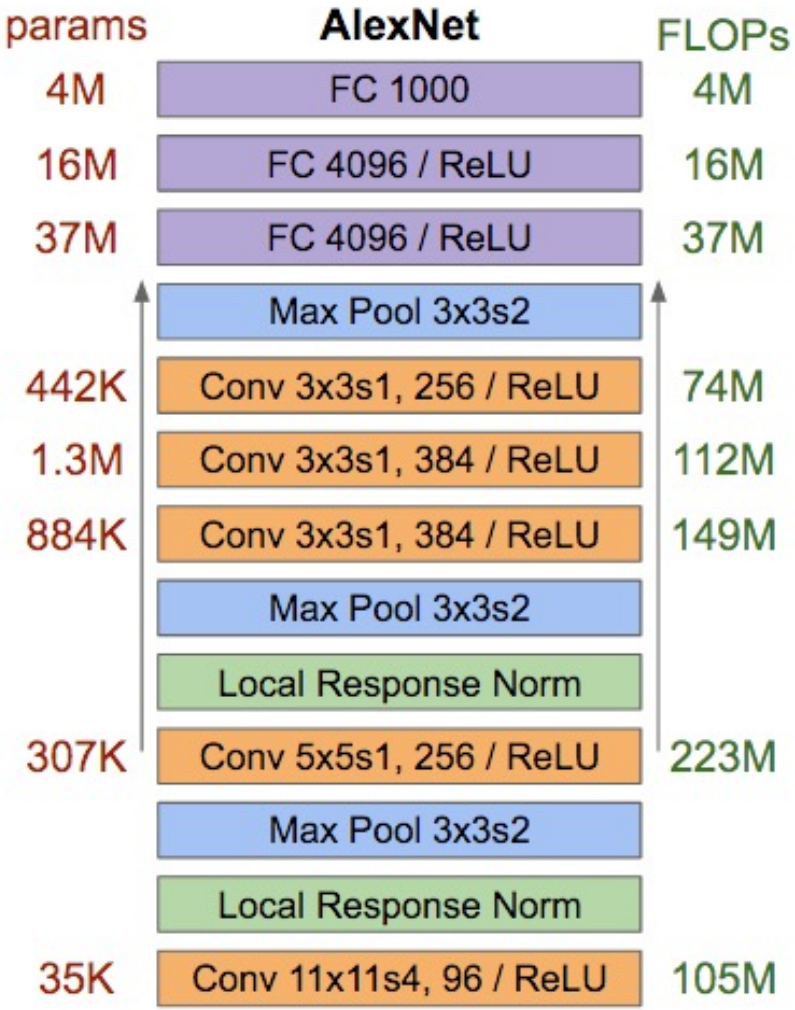
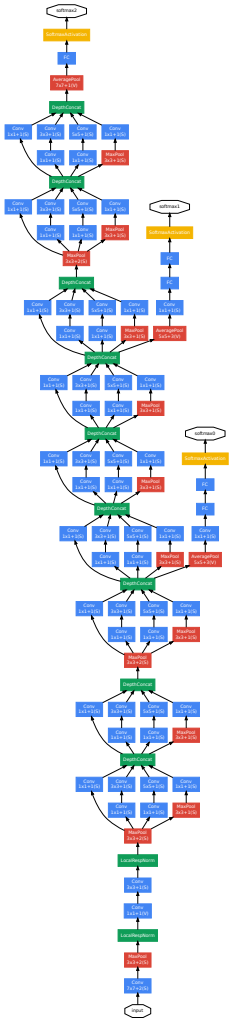
Convolutional Neural Networks



AlexNet: ImageNet Classification with Deep Convolutional Neural Networks



GoogLeNet: Going deeper with convolutions



Motivation

How to design neural network architectures automatically?

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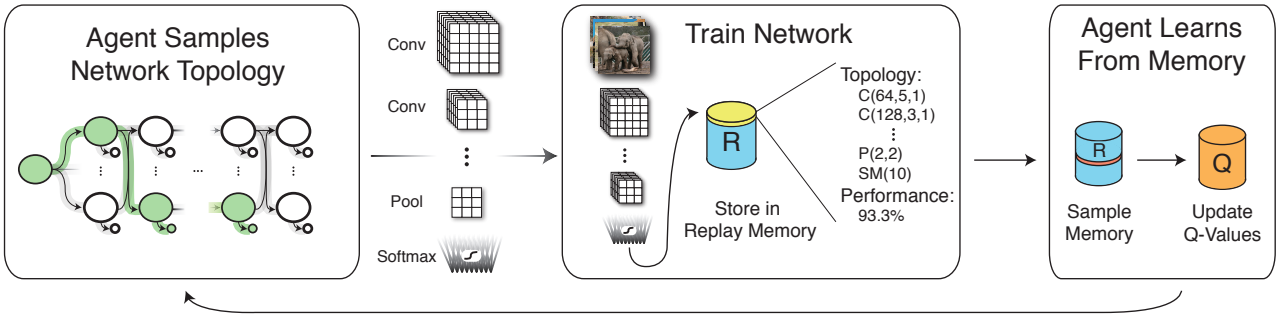
Related Work

Categorized by search methods:

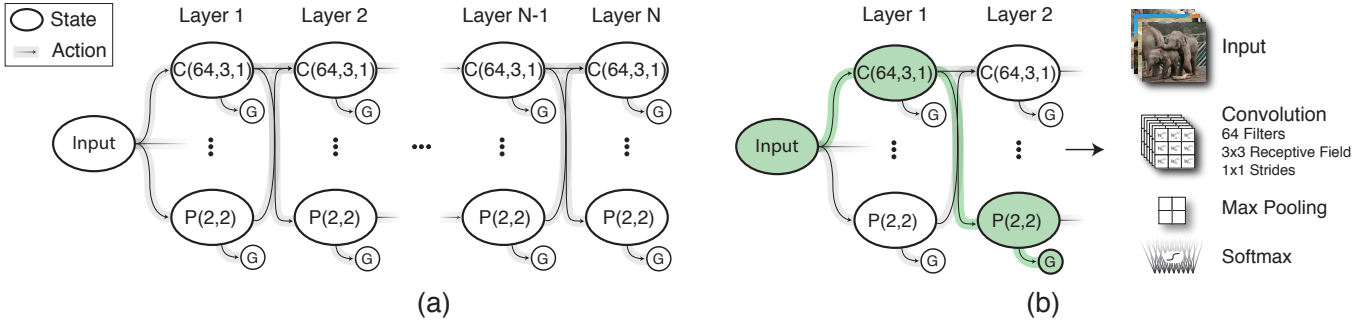
- Reinforcement learning (RL)
- Evolutionary algorithms (EA)
- Sequential model-based optimization (SMBO)
- Gradient ascent (GA)

The Workflow of MetaQNN

Framework:



Generation:



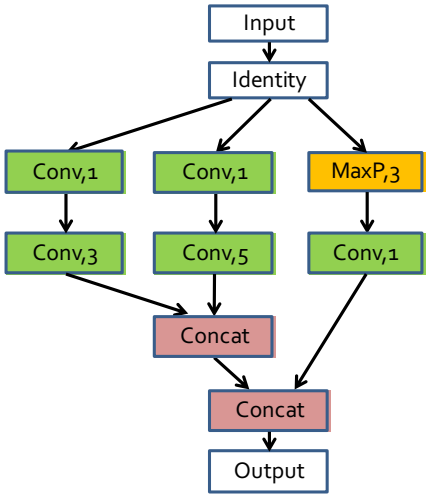
Experimental State Space

Layer Type	Layer Parameters	Parameter Values
Convolution (C)	$i \sim$ Layer depth $f \sim$ Receptive field size $\ell \sim$ Stride $d \sim$ # receptive fields $n \sim$ Representation size	< 12 Square. $\in \{1, 3, 5\}$ Square. Always equal to 1 $\in \{64, 128, 256, 512\}$ $\in \{(\infty, 8], (8, 4], (4, 1]\}$
Pooling (P)	$i \sim$ Layer depth $(f, \ell) \sim$ (Receptive field size, Strides) $n \sim$ Representation size	< 12 Square. $\in \{(5, 3), (3, 2), (2, 2)\}$ $\in \{(\infty, 8], (8, 4]$ and $(4, 1]\}$
Fully Connected (FC)	$i \sim$ Layer depth $n \sim$ # consecutive FC layers $d \sim$ # neurons	< 12 < 3 $\in \{512, 256, 128\}$
Termination State	$s \sim$ Previous State $t \sim$ Type	Global Avg. Pooling/Softmax

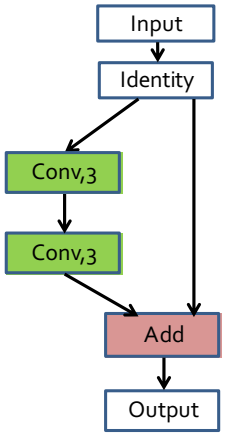
10 GPUs $\times$ 9-10 days on each dataset

Advanced Network Structure Code Space

Name	Index	Type	Kernel Size	Pred1	Pred2
Convolution	**T**	1	1, 3, 5	**K**	0
Max Pooling	**T**	2	1, 3	**K**	0
Average Pooling	**T**	3	1, 3	**K**	0
Identity	**T**	4	0	**K**	0
Elemental Add	**T**	5	0	**K**	**K**
Concat	**T**	6	0	**K**	**K**
Terminal	**T**	7	0	0	0



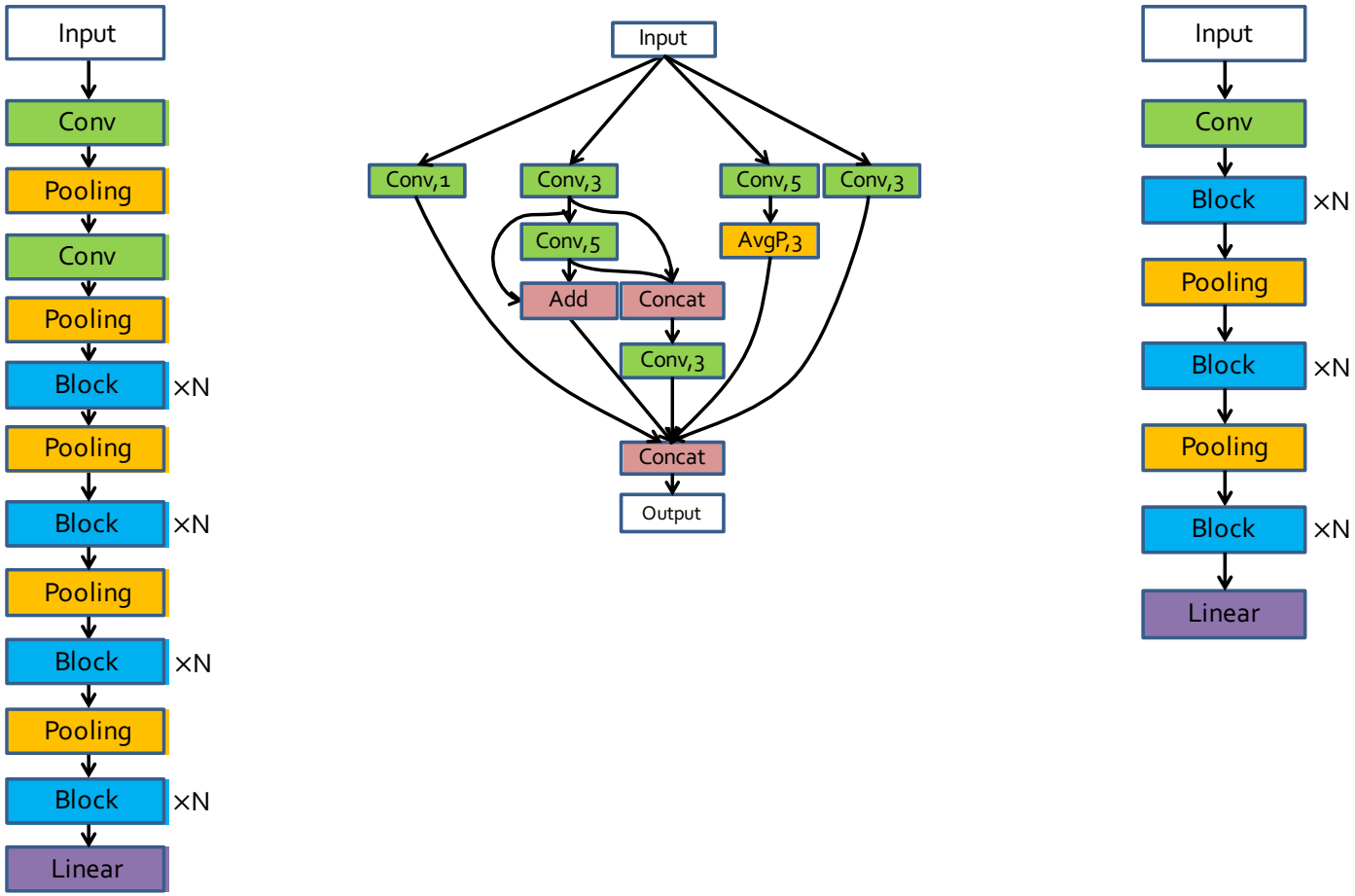
Codes = [(1,4,0,0,0), (2,1,1,1,0), (3,1,3,2,0), (4,1,1,1,0), (5,1,5,4,0), (6,6,0,3,5), (7,2,3,1,0), (8,1,1,7,0), (9,6,0,6,8), (10,7,0,0,0)]



Codes = [(1,4,0,0,0), (2,1,3,1,0), (3,1,3,2,0), (4,5,0,1,3), (5,7,0,0,0)]

32 GPUs $\times$ 3 days

Result of BlockQNN



Using a RNN as Architecture Generator

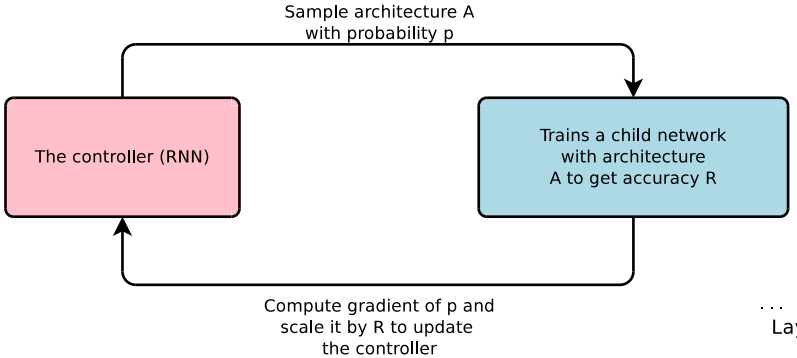


Figure 1: An overview of Neural Architecture Search.

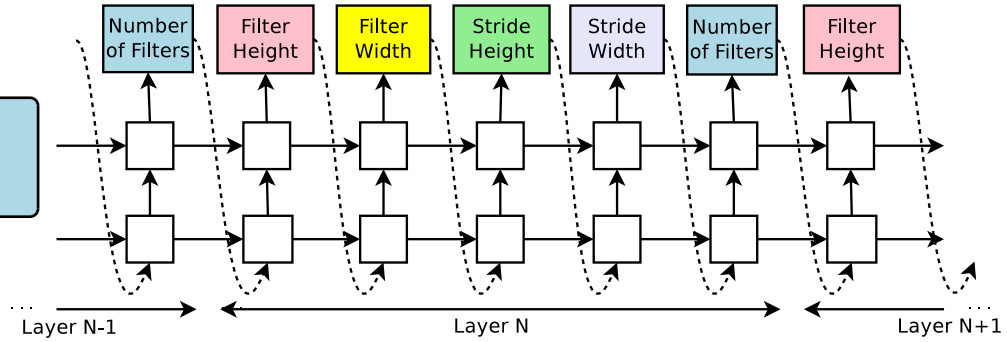
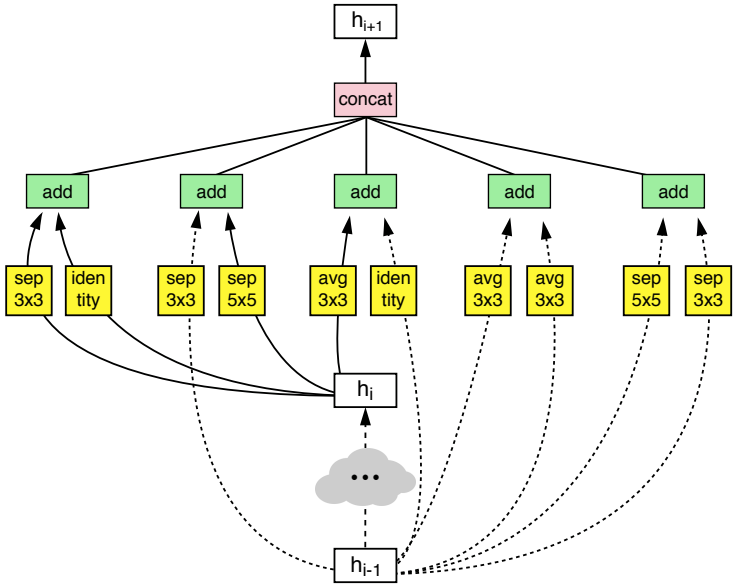


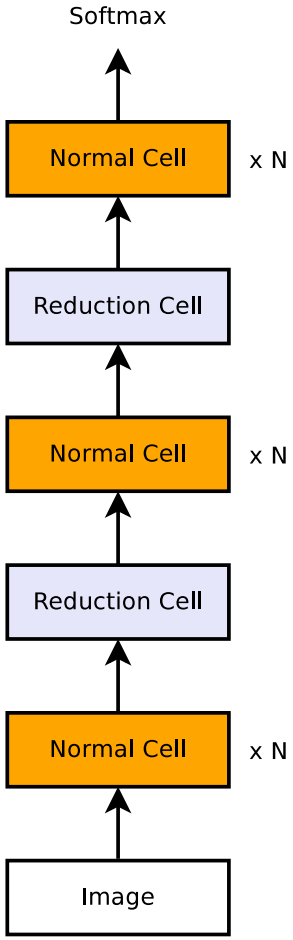
Figure 2: How the controller recurrent neural network samples a simple convolutional network.

800 GPUs $\times$ 28 days

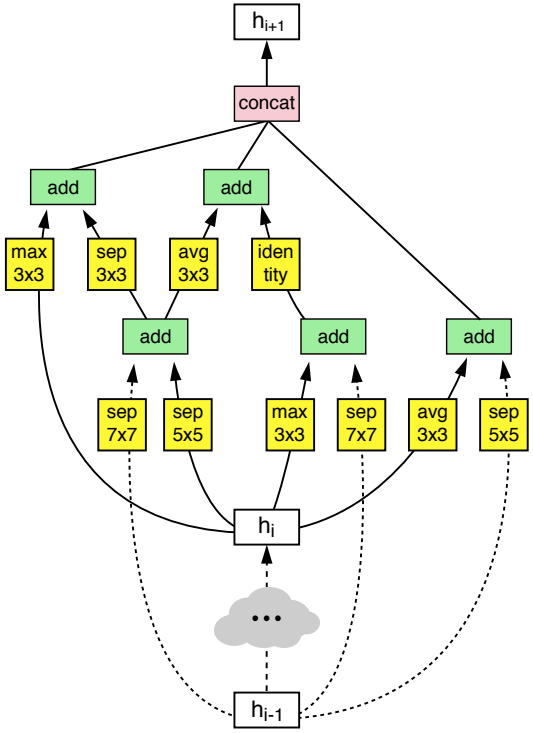
NASNet on CIFAR-10



Normal Cell



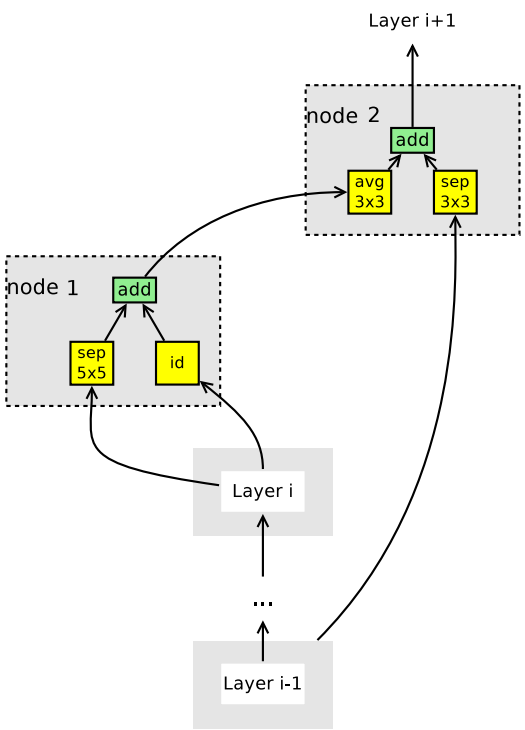
CIFAR10
Architecture



Reduction Cell

Cell Topologies of NASNet

The cell can be represented by a DAG consisting of B nodes. Each node is a mapping from 2 input tensors to 1 output tensor and can be specified as a 4-tuple.

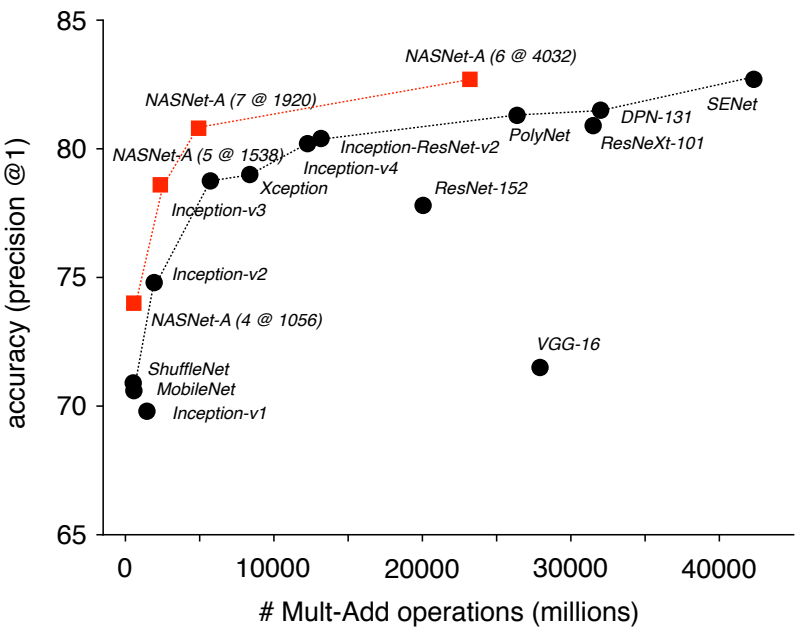


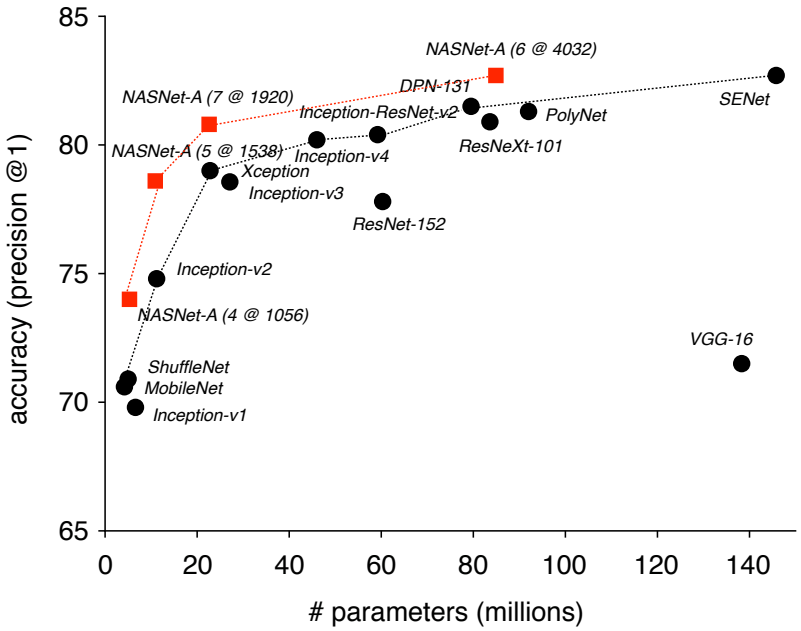
Definition

Node tuple in CNN cell: $< I_1, I_2, O_1, O_2 >$

- $I_1, I_2 \in \mathcal{I}_b$ specifies the inputs to the node
- $O_1, O_2 \in \mathcal{O}$ specifies the operation to apply to input I_i
- $\mathcal{I}_b = \{H_1^c, ..., H_{b-1}^c, H_B^{c-1}, H_B^{c-2}\}$ is the set of outputs of all previous nodes in this cell and the outputs of two previous cell
- $\mathcal{O}$ is the set of 11 functions, each of which operates on a single tensor

Performance on ImageNet





500 GPUs $\times$ 4 days

Parameter Sharing Technology

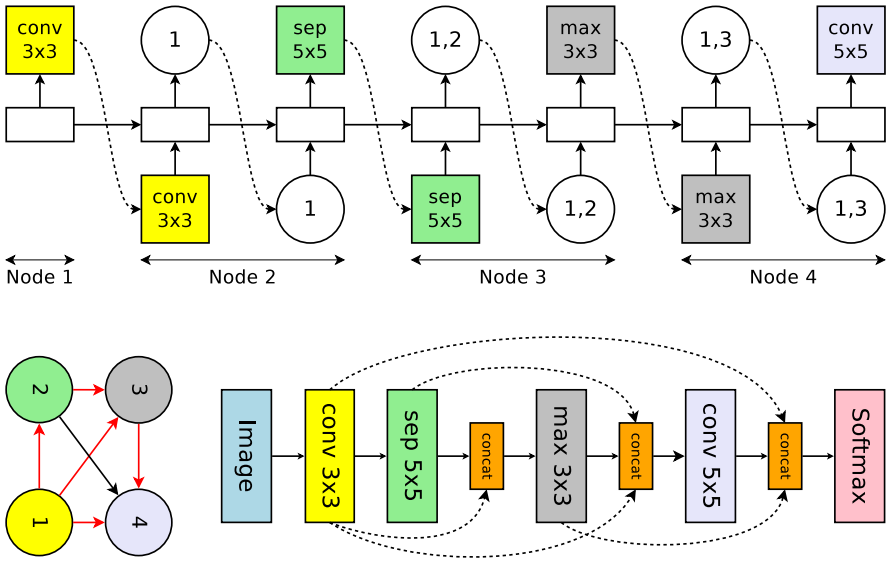


Figure 1. An example run of a recurrent cell in our search space with 4 computational nodes, which represent 4 layers in a con-volutional network. *Top:* The output of the controller RNN. *Bot-tom Left:* The computational DAG corresponding to the network's architecture. Red arrows denote the active computational paths. *Bottom Right:* The complete network. Dotted arrows denote skip connections.

1 GPU $\times$ 0.45 day

Formally, an architecture is defined by the representation *(G,* $\mathbf{o}$*)*, consisting of two ingredients:

- A set of available operations $\mathbf{o} = \{o_1, o_2, ...\}$
- An adjacency matrix G specifying the neural network graph of operations, where $G_{ij} = k$ means that the k-th operation o_k is to be placed between nodes i and j

The architecture is obtained by assembling operations $\mathbf{o}$ according to the adjacency matrix G:

$$arch = assemble(G, \mathbf{o})$$

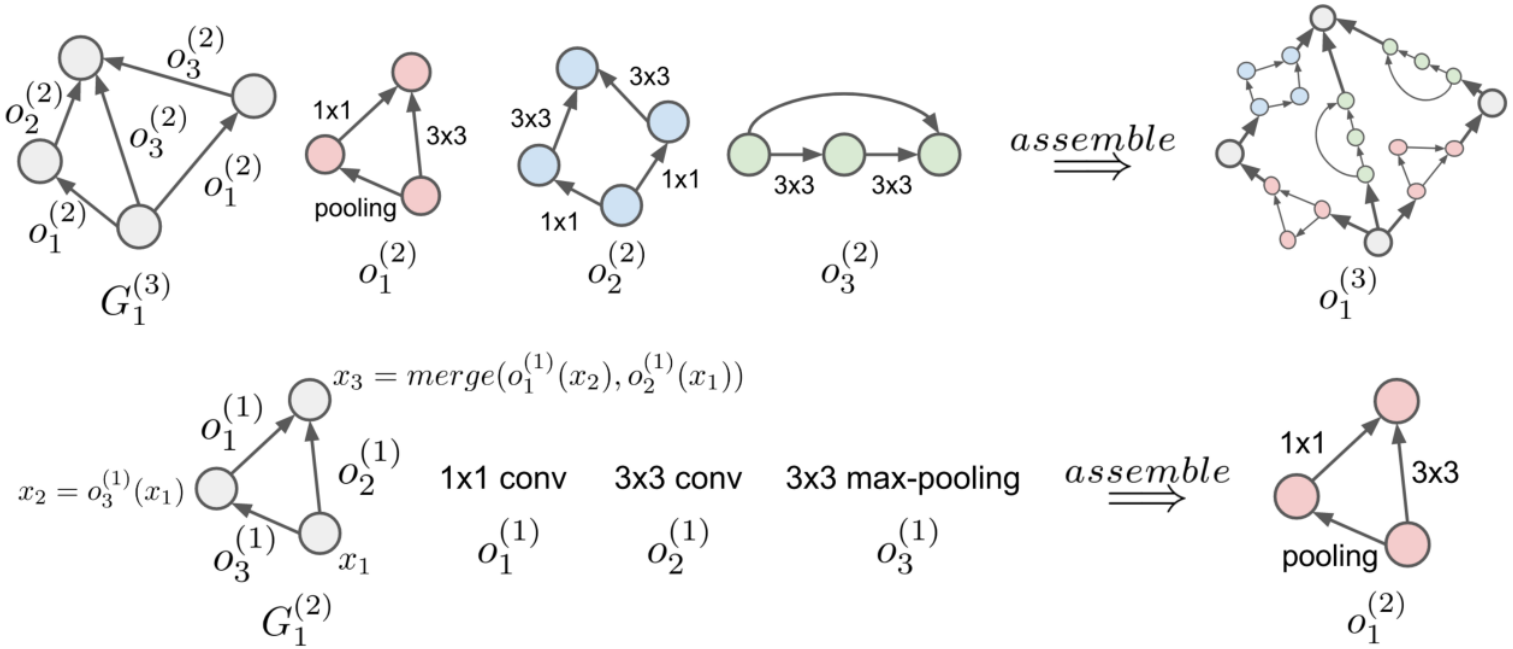


Figure 1: An example of a three-level hierarchical architecture representation. The bottom row shows how level-1 primitive operations $o_1^{(1)}, o_2^{(1)}, o_3^{(1)}$ are assembled into a level-2 motif $o_1^{(2)}$. The top row shows how level-2 motifs $o_1^{(2)}, o_2^{(2)}, o_3^{(2)}$ are then assembled into a level-3 motif $o_1^{(3)}$.

200 GPUs $\times$ 1.5 days

Algorithms

Algorithm 1: ASYNCEVO Asynchronous Evolution (Controller)

Input: Data queue $\mathcal{Q}$ containing initial genotypes; Memory table $\mathcal{M}$ recording evaluated genotypes and their fitness.

while *True* **do**
  if HASIDLEWORKER() **then**
    $genotype \leftarrow$ ASYNCTOURNAMENTSELECT($\mathcal{M}$)
    $genotype' \leftarrow$ MUTATE($genotype$)
    $\mathcal{Q} \leftarrow \mathcal{Q} \cup genotype'$

Algorithm 2: ASYNCEVO Asynchronous Evolution (Worker)

Input: Training set $\mathcal{T}$, validation set $\mathcal{V}$; Shared memory table $\mathcal{M}$ and data queue $\mathcal{Q}$.

while *True* **do**
  if $|\mathcal{Q}| > 0$ **then**
    $genotype \leftarrow \mathcal{Q}$.pop()
    $arch \leftarrow$ ASSEMBLE($genotype$)
    $model \leftarrow$ TRAIN($arch, \mathcal{T}$)
    $fitness \leftarrow$ EVALUATE($model, \mathcal{V}$)
    $\mathcal{M} \leftarrow \mathcal{M} \cup (genotype, fitness)$

Illustration of PNAS

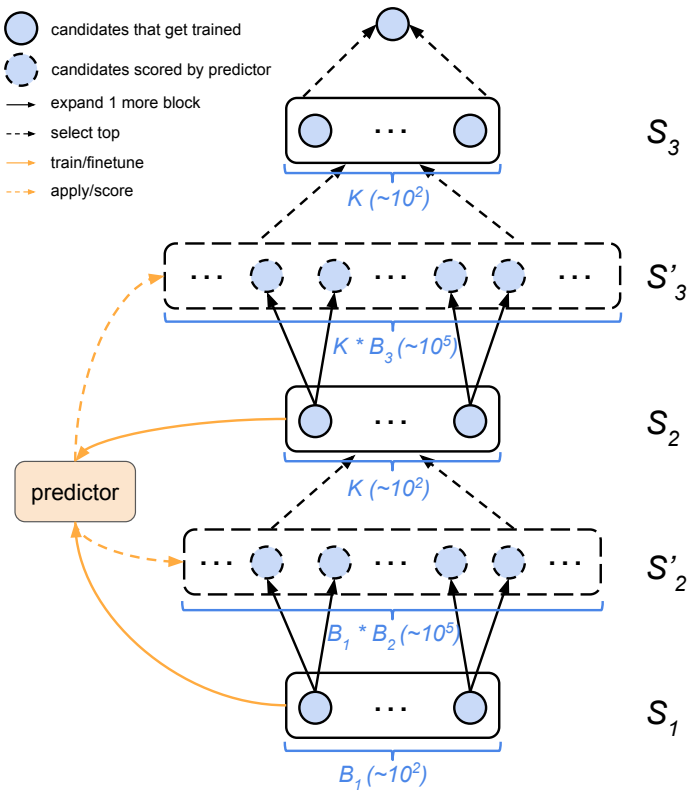
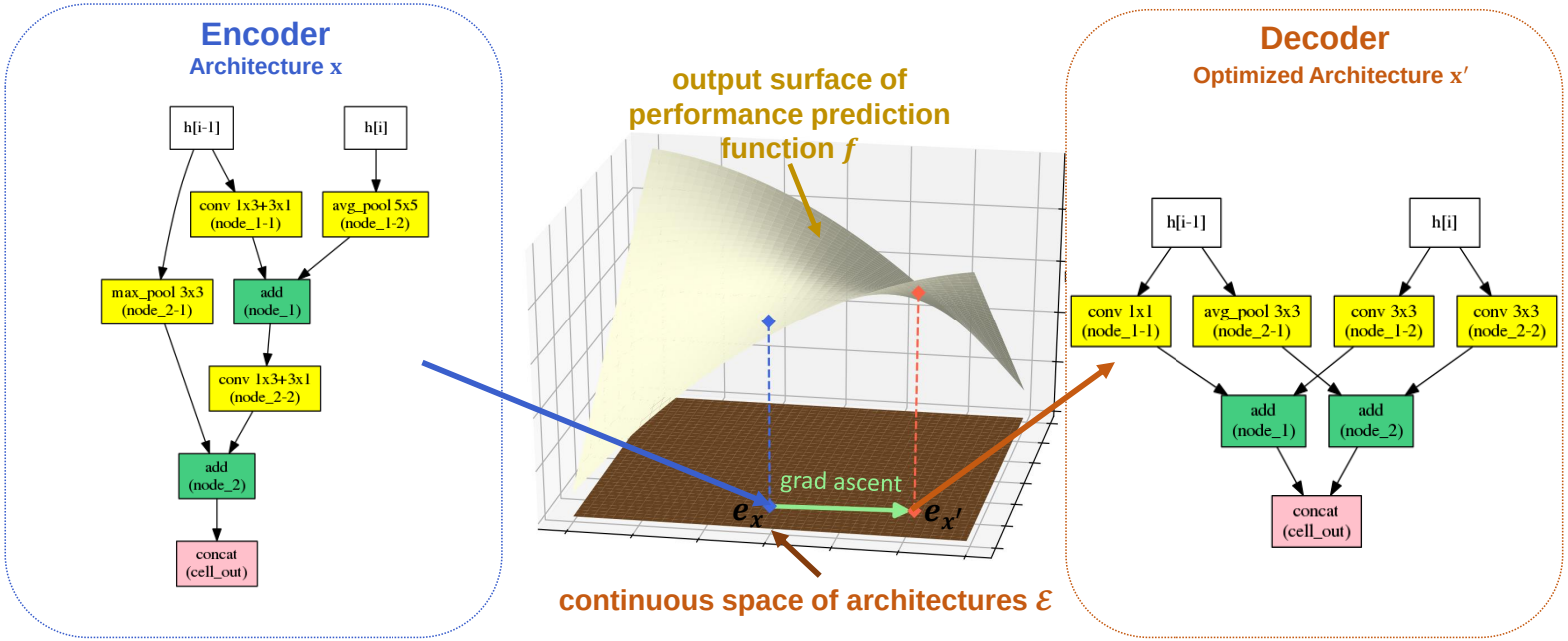


Fig. 1. Illustration of the PNAS search procedure when the maximum number of blocks is $B = 3$. Here $\mathcal{S}_b$ represents the set of candidate cells with b blocks. We start by considering all cells with 1 block, $\mathcal{S}_1 = \mathcal{B}_1$; we train and evaluate all of these cells, and update the predictor. At iteration 2, we expand each of the cells in $\mathcal{S}_1$ to get all cells with 2 blocks, $\mathcal{S}'_2 = \mathcal{B}_{1:2}$; we predict their scores, pick the top K to get $\mathcal{S}_2$, train and evaluate them, and update the predictor. At iteration 3, we expand each of the cells in $\mathcal{S}_2$, to get a subset of cells with 3 blocks, $\mathcal{S}'_3 \subseteq \mathcal{B}_{1:3}$; we predict their scores, pick the top K to get $\mathcal{S}_3$, train and evaluate them, and return the winner. $B_b = |\mathcal{B}_b|$ is the number of possible blocks at level b and K is the beam size (number of models we train and evaluate per level of the search tree).

225 GPU days

The Overall Framework of NAO



- The original architecture x is mapped to continuous representation e_x via encoder network
- e_x is optimized into $e_{x'}$ via maximizing the output of performance predictor f using gradient ascent
- $e_{x'}$ is transformed into a new architecture x' using the decoder network

Detailed Algorithm

Algorithm 1 Neural Architecture Optimization

Input: Initial candidate architectures set X to train NAO model. Initial architectures set to be evaluated denoted as $X_{eval} = X$. Performances of architectures $S = \emptyset$. Number of seed architectures K. Step size η. Number of optimization iterations L.

for $l = 1, \cdots, L$ **do**

Train each architecture $x \in X_{eval}$ and evaluate it to obtain the dev set performances $S_{eval} = \{s_x\}, \forall x \in X_{eval}$. Enlarge S: $S = S \bigcup S_{eval}$.

Train encoder E, performance predictor f and decoder D by minimizing Eqn.(1), using X and S.

Pick K architectures with top K performances among X, forming the set of seed architectures X_{seed}.

For $x \in X_{seed}$, obtain a better representation $e_{x'}$ from $e_{x'}$ using Eqn. (2), based on encoder E and performance predictor f. Denote the set of enhanced representations as $E' = \{e_{x'}\}$.

Decode each x' from $e_{x'}$ using decoder, set X_{eval} as the set of new architectures decoded out: $X_{eval} = \{D(e_{x'}), \forall e_{x'} \in E'\}$. Enlarge X as $X = X \bigcup X_{eval}$.

end for

Output: The architecture within X with the best performance

200 GPUs $\times$ 1 day & 0.4 GPU day (+weight sharing)

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Key Issues in NAS

- Design a high precision performance predictor
- Construct efficient architecture space
- Accelerate search process and improve sampling efficiency
- Develop multiple objective/reward

Google Cloud AutoML

- https://cloud.google.com/automl/
- Vision, Natural Language and Translation

Thank you for your attention!